

Original four-model pathway robustness checks

	ssGSEA	GSEA	Held-out	3 seeds	Composition
Thymus: DNA repair	+	+	+	+	+
Thymus: Extracellular matrix organization	+	+	+	+	-
Thymus: Innate TLR signaling	+	+	+	-	+
Thymus: Lymphoid-stromal interactions	-	-	+	+	+
Thymus: Mitotic cell cycle	+	+	+	-	+
Thymus: RHOA cytoskeletal cycle	+	+	+	+	+
Thymus: T-cell receptor signaling	+	+	+	+	-
Skin: Cell-cell junction organization	-	+	+	+	+
Skin: Chromatin-modifying enzymes	+	+	+	+	+
Skin: DNA repair	+	+	+	+	+
Skin: Gap-junction trafficking	+	+	+	-	+
Skin: Hedgehog signaling	+	+	+	+	+
Skin: Keratinization	+	+	+	-	+
Skin: Phase II detoxification	+	+	+	-	+
Skin: Sphingolipid metabolism	+	+	+	+	+
Liver: Cytochrome P450	+	+	-	+	+
Liver: Extracellular matrix organization	+	+	+	+	-
Liver: Glutathione conjugation	+	+	-	-	+
Liver: MHC class II antigen presentation	+	+	+	+	+
Liver: Regulation of insulin secretion	-	+	+	-	+
Liver: Rho-family GTPase cycle	+	+	+	-	+
Liver: T-cell receptor signaling	+	+	+	+	+
Soleus: Cytokine signaling	-	-	+	-	+
Soleus: DNA repair	+	+	+	+	-
Soleus: Extracellular matrix degradation	-	-	+	-	-
Soleus: Fatty-acid metabolism	+	+	-	+	-
Soleus: Glycosaminoglycan metabolism	+	+	+	+	-
Soleus: Immune-system signaling	-	-	-	+	-
Soleus: Striated-muscle contraction	+	+	-	-	-